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# Scalability-Aware Benchmarking of Graph Anomaly Detection on Financial and Cyber Networks: Evaluated Population, Alert Budgets, and the Exact Limits of Decoupled GNNs

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**Abstract**

Originally developed for fraud detection on Ethereum, BSC, and Polygon, the Decoupled Local-to-Global Graph Neural Network (DLG-GNN) requires evaluation beyond its source domain. We benchmark five established graph anomaly detectors alongside DLG-Base (without local augmentation) and DLG-Aug (with pretrained local representations) across 13 primary graphs and the external LANL-RedTeam network using a PyGOD-compatible implementation. Across five seeds, 84 of 91 primary model–graph pairs completed successfully (420 primary and 35 external runs); unsupported cases were reported explicitly. DLG-Aug improved average precision (AP) over DLG-Base on Elliptic by 0.0350 but reduced it on Reddit-Syn by 0.0656 and underperformed on LANL-RedTeam. On each of the three original blockchain domains, at least one established comparator achieved higher mean AP than DLG-Aug. We evaluate AP against test-set anomaly prevalence and report precision and recall at 1% and 5% alert budgets. Exact sparse reconstruction avoids dense quadratic matrix materialization for compatible decoders, with numerical equivalence tests; row-block AnomalyDAE execution still requires quadratic arithmetic. These results provide conditional usage guidance and documented computational limits for reproducing and reusing DLG-GNN.

**Keywords:** graph anomaly detection; graph neural networks; financial fraud detection; alert budget; exact sparse reconstruction; reproducible benchmarking; local-to-global learning

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## 1. Introduction

Graph anomaly detection (GAD) has emerged as a cornerstone of modern machine learning for financial fraud investigation, blockchain anti-money laundering (AML), and enterprise cybersecurity monitoring [1–3]. By modeling transactions, contracts, and authentication events as interconnected graphs $G = (V, E, X)$, graph neural networks (GNNs) can capture subtle topological irregularities and feature discrepancies that evade tabular screening.

However, moving GAD from idealized academic benchmarks to operational deployment reveals fundamental challenges that conventional evaluation protocols frequently obscure:

1. **Failure-Blind Reporting and Hardware Infeasibility:** Standard leaderboards often omit unsupported cells when models suffer out-of-memory (OOM) crashes or runtime

timeouts on web-scale or multi-million-node graphs, or they quietly replace the full-graph objective with localized neighborhood sampling without verifying whether score semantics are preserved.

2. **Imbalance Disconnect and Below-Random Behavior:** In real-world financial fraud, anomaly prevalence is frequently below 2% or 0.5%. In these extreme regimes, high area under the ROC curve (ROC-AUC) can be misleading. More critically, unsupervised reconstruction models often exhibit empirical ROC-AUC scores falling well below the 0.50 random baseline on financial transaction networks such as Elliptic under standard score conventions. While topological characteristics such as high-degree licit transaction hubs have been hypothesized to incur disproportionate reconstruction errors, the absence of archived per-node score arrays precludes causal attribution; we document this below-random behavior as an authentic empirical finding without heuristic sign inversion.
3. **Alert-Budget Constraints:** Security operations centers (SOCs) and fraud investigation teams cannot inspect thousands of suspicious accounts; they operate under strict investigative triage budgets (e.g., auditing the top-1% or top-5% flagged alerts). Ranking metrics alone do not tell practitioners whether a detector provides actionable precision at the head of the flagged list.
4. **Conditional Utility of Architecture Augmentation:** Pretraining local graph representations and fusing them with global message passing (e.g., Decoupled Local-to-Global architectures, DLG [4]) has demonstrated strong performance in specific cryptocurrency domains, but its comparative utility across diverse topologies remains unverified under frozen, reproducible protocols.

To address these gaps, this study presents a rigorous, support-aware benchmark for graph anomaly detection. We freeze seven representative detector configurations across 13 primary graph datasets and one external cybersecurity dataset, executing five random seeds per combination. Rather than claiming universal superiority for any single model, our benchmark contributes:

1. **A Support-Aware Evaluation Protocol:** We explicitly distinguish between successful execution (`SUPPORTED_EXACT`) and unsupported pairs across three verified failure categories (`NO_APPROVED_RAW_RUN`, `OBSERVED_HISTORICAL_CUDA_OOM`, and `OBSERVED_CUDA_OOM`), evaluating exactly 84 of 91 primary pairs (420 successful primary runs plus 35 external LANL runs; 455 total runs) without heuristic score imputation.
2. **Audited Predictive Quality vs. Evaluation Prevalence and Alert Budgets:** We pair Average Precision (AP) with explicit evaluation prevalence references and evaluate triage precision and recall at fixed 1% and 5% alert budgets across financial and cyber graphs.
3. **Rigorous Diagnosis of Below-Random Inversion:** We conduct an in-depth score, label, and mask audit of Elliptic and DGraphFin, demonstrating that below-0.50 ROC-AUC values reflect authentic empirical behavior under standard detector orientation rather than software defects, while exploring informed topological hypotheses.
4. **Exact Closed-Form Decoupling Without Dense Materialization:** We mathematically derive and qualify closed-form Gram-matrix reconstructions that eliminate $N \times N$ dense adjacency storage, providing exact mathematical equivalence to dense loss evaluations for unsquashed dot-product decoders.
5. **Application-Facing Decision Matrix:** We synthesize our findings into an actionable practitioner decision guide that maps graph density, feature availability, and alert budgets to optimal detector selection.

## 2. Related Work

### 2.1. Unsupervised Graph Anomaly Detection

Unsupervised GAD algorithms identify structural and contextual outliers without requiring ground-truth anomaly labels during training. DOMINANT [5] pioneered graph autoencoders by jointly minimizing attribute reconstruction error and dense adjacency reconstruction loss. AnomalyDAE [6] employs dual autoencoders with cross-modal attention between structure and features. Contrastive methods such as CoLA [7] and CONAD [8] train local-to-subgraph mutual information estimators, scoring nodes by contrastive disagreement. However, due to PyGOD's CONAD implementation requiring full perturbed graph generation whose reproducibility was archived only as a preliminary diagnostic baseline without approved multi-seed canonical runs, CONAD is evaluated strictly as a diagnostic comparator rather than a primary benchmark model. However, due to PyGOD's CONAD implementation requiring full perturbed graph generation whose reproducibility was archived only as a preliminary diagnostic baseline without approved multi-seed canonical runs, CONAD is evaluated strictly as a diagnostic comparator rather than a primary benchmark model. One-class models such as OCGNN [9] optimize a hypersphere enclosing normal representations, while GADNR [10] reconstructs neighborhood feature distributions via Gaussian mixture modeling.

Comprehensive surveys [1,2] and benchmark initiatives such as PyGOD [11], BOND [12], and GADBench [13] have standardized implementations across small and medium graphs. Recent specialized architectures, including PREM [14], TAM [15], GADAM [16], and ADAG [17], explore graph prompt learning, truncation, and adaptive augmentation. Recent investigations, such as GAD-in-the-Wild [18], have highlighted performance drops under realistic attribute corruption and low class prevalence. In cybersecurity, Dilini *et al.* [3] comprehensively reviewed graph-based anomaly detection methods for network intrusion detection, demonstrating that real-world authentication and flow graphs require specialized topological modeling. Our work builds upon this foundation by establishing hardware-aware execution boundaries and providing closed-form exact loss formulations for full-graph training.

### 2.2. Imbalance-Aware Fraud Learning and Signed Networks

Financial transaction graphs are characterized by extreme class skew and heterophily. Semi-supervised and supervised approaches such as PC-GNN [19], GraphSMOTE [20], and Dual-Augment GNN [21] resample fraud distributions or penalize minority-class misclassification. In signed transaction networks, edge weights denote trust or distrust, as formalized by Kumar *et al.* [22] in Bitcoin trust networks. For smart contract platforms, Jiang and Tsai [23] proposed MVCG-SPS, a multi-view contrastive graph neural network for smart Ponzi scheme detection on Ethereum, demonstrating the effectiveness of decoupled relational perspectives. Because supervised fraud detectors require verified labels that are rarely available in early-stage forensics, this benchmark focuses primarily on unsupervised detectors, but incorporates alert-budget precision and recall to reflect practical operational requirements.

### 2.3. Scalable Graph Learning and Sampling Limits

To overcome the $O(N^2)$ memory bottleneck of dense adjacency matrices and full-graph convolutions, large-scale GNN frameworks employ neighborhood sampling (GraphSAGE [24]), layer sampling (FastGCN [25], LADIES [26]), or graph partitioning (Cluster-GCN [27], GraphSAINT [28]). While effective for node classification, sampling alters the empirical structural objective and can introduce variance that distorts reconstruction-based anomaly

scores. Our approach instead retains the exact full-graph objective while leveraging algebraic reformulations to achieve scalable execution without sampling approximations.

### *2.4. Benchmark Methodology and Evaluation Rigor*

Graph model rankings often exhibit strong sensitivity to split generation, hyperparameter choices, and random seeds [29,30]. OGB [31] and controlled benchmarking efforts [32] have emphasized the necessity of frozen data splits and transparent failure reporting. Crucially, Heidrich *et al.* [33] established a systematic evaluation method for graph-derived signals, highlighting that multi-seed statistical significance and explicit dataset provenance are paramount when assessing model utility. Our protocol adheres to these principles by executing identical seed splits across all models, reporting non-parametric within-block permutation statistics, and leaving unsupported cells unscored.

## 3. Decoupled GNNs and Exact Gram Sparse Decoders

### *3.1. Evaluated Detector Architectures*

We evaluate seven functioning anomaly detector configurations spanning autoencoder, contrastive, one-class, and decoupled architectural families:

1. **DOMINANT** [5]: GCN autoencoder reconstructing node attributes $X$ and dense adjacency $A$.
2. **AnomalyDAE** [6]: Dual autoencoder decoupling feature and structural embeddings via cross-modal attention.
3. **CoLA** [7]: Contrastive self-supervised model evaluating local sub-graph agreement.
4. **GADNR** [10]: Neighborhood distribution reconstruction with Gaussian mixture modeling.
5. **OCGNN** [9]: One-class classification optimizing a hypersphere enclosing normal representations.
6. **DLG-Base** [4]: Standard GCN autoencoder employing exact sparse Gram loss evaluation.
7. **DLG-Aug** [4]: Decoupled Local-to-Global model combining finite-receptive local GCN pre-training with augmented global relational propagation.

### *3.2. Decoupled Local-to-Global Architecture*

The overall workflow of the Decoupled Local-to-Global (DLG) architecture is illustrated in Figure 1. The framework encompasses two distinct variants: *DLG-Base* and *DLG-Aug* (DLG-Full).

**DLG-Base Module:** DLGBase couples a Level 1 local encoder and a Level 2 global encoder sequentially. Given node attributes $X \in \mathbb{R}^{N\times D}$ and edge index $\mathcal{E}$, the local encoder generates neighborhood-aware representation $Z_{\text{local}} = \text{GCN}_1(X, \mathcal{E})$, which directly feeds the global encoder $Z_{\text{global}} = \text{GCN}_2(Z_{\text{local}}, \mathcal{E})$. A learnable scalar parameter $\alpha$ fuses both representations: $Z_{\text{fused}} = \sigma(\alpha)Z_{\text{local}} + (1-\sigma(\alpha))Z_{\text{global}}$. The attribute decoder reconstructs $\hat{X} = \text{GCN}_{\text{dec}}(Z_{\text{fused}}, \mathcal{E})$, and the structure decoder reconstructs adjacency via dot product $\hat{S} = Z_{\text{fused}}Z_{\text{fused}}^{\top}$.

**DLG-Aug (DLG-Full Pipeline):** In contrast to DLG-Base, DLG-Aug decouples the local and global stages into a two-phase training protocol. In Stage 1 (Local Pre-training), a separate 2-layer GCN autoencoder ($k = 2$ message-passing receptive field) is trained on input attributes $X$ by minimizing mean squared feature reconstruction error $\|X - \hat{X}_{\text{L1}}\|_F^2$. Crucially, this pre-training operates in full-graph forward passes without explicit ego-net subgraph extraction. Once trained, its parameters are discarded, and the extracted per-node embeddings $Z_{\text{local}} \in \mathbb{R}^{N\times d}$ are detached and frozen. In Stage 2 (Global Encoding), the input feature matrix is augmented via concatenation $\tilde{X} = [X \parallel Z_{\text{local}}] \in \mathbb{R}^{N\times(D+d)}$. The

global GCN encoder processes $\tilde{X}$ over $\mathcal{E}$ to produce latent embedding $Z \in \mathbb{R}^{N \times d}$. The attribute decoder then reconstructs the *original* attributes $\hat{X} \in \mathbb{R}^{N \times D}$ (not the augmented $\tilde{X}$), while the structure decoder computes $\hat{S} = ZZ^\top$.

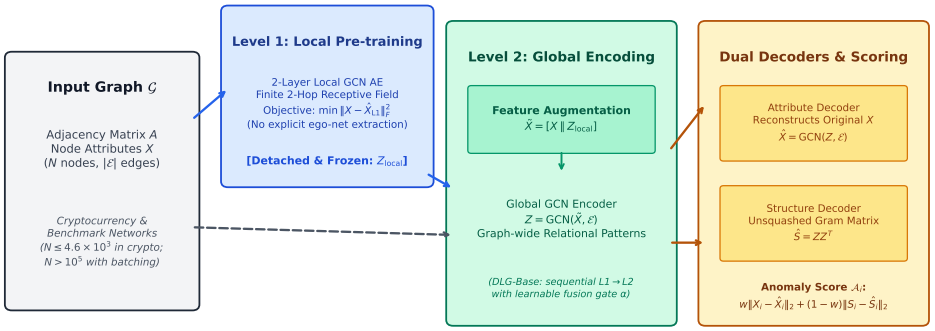



**Figure 1.** Decoupled Local-to-Global (DLG) anomaly detection architecture. In Level 1, a local GCN feature autoencoder pretrains on full-graph attributes using a finite 2-hop message-passing receptive field (without explicit ego-subgraph induction) and extracts frozen local representation $\mathbf{Z}_{\text{local}}$. In Level 2, the global relational encoder processes the augmented representation $[\mathbf{X} \,\|\, \mathbf{Z}_{\text{local}}]$ across graph topology $\mathcal{E}$. The attribute decoder reconstructs original attributes $\mathbf{X}$, while the structure decoder computes the unsquashed Gram matrix $\hat{S} = \mathbf{Z}\mathbf{Z}^\top$. Per-node anomaly scores combine Euclidean attribute and structural reconstruction residuals.

### *3.3. Closed-Form Exact Gram Decoders*

A primary source of memory failure in structural autoencoders (e.g., DOMINANT, DLG-Base, DLG-Aug) is the evaluation of the squared Frobenius reconstruction error against adjacency matrix $A \in \{0,1\}^{N \times N}$:

$$\mathcal{L}_{\text{struct}} = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \left( A_{ij} - \mathbf{z}_i^\top \mathbf{z}_j \right)^2, \tag{1}$$

where $\mathbf{z}_i \in \mathbb{R}^d$ denotes the latent embedding of node $i$.

In standard implementations, direct computation requires materializing the dense inner product matrix $\mathbf{Z}\mathbf{Z}^\top \in \mathbb{R}^{N \times N}$. For single-precision float32 data, a single contiguous $N \times N$ matrix requires $4N^2$ bytes, which corresponds to 10.0 GB (9.31 GiB) at $N = 50,000$ and 160.0 GB (149.01 GiB) at $N = 200,000$. Materializing two such dense matrices simultaneously (the target adjacency $A$ and the predicted reconstruction $\mathbf{Z}\mathbf{Z}^\top$) requires $8N^2$ bytes, consuming approximately 20.0 GB and 320.0 GB in decimal gigabytes, respectively, before autograd computational graph allocations and workspace overhead. This dense materialization rapidly exhausts GPU device memory.

For unsquashed dot-product decoders (the default setting in PyGOD DOMINANT with `sigmoid_s=False`), we expand Equation (1) algebraically:

$$\begin{aligned} \sum_{j=1}^{N} \left( A_{ij} - \mathbf{z}_i^\top \mathbf{z}_j \right)^2 &= \sum_{j=1}^{N} A_{ij}^2 - 2\mathbf{z}_i^\top \sum_{j=1}^{N} A_{ij} \mathbf{z}_j + \sum_{j=1}^{N} (\mathbf{z}_i^\top \mathbf{z}_j)^2 \\ &= \deg(i) - 2\mathbf{z}_i^\top (A\mathbf{Z})_i + \mathbf{z}_i^\top \left( \mathbf{Z}^\top \mathbf{Z} \right) \mathbf{z}_i, \end{aligned} \tag{2}$$

where $\deg(i) = \sum_j A_{ij}^2 = \sum_j A_{ij}$ for unweighted binary graphs. At inference, following standard PyGOD semantics, the per-node structural anomaly score $\mathcal{A}_{\text{struct},i}$ is the Euclidean $L_2$ norm of the reconstruction residual vector: $\mathcal{A}_{\text{struct},i} = \|A_{i,:} - \hat{S}_{i,:}\|_2 =$

$\sqrt{\deg(i) - 2\mathbf{z}_i^\top (A\mathbf{Z})_i + \mathbf{z}_i^\top (\mathbf{Z}^\top \mathbf{Z})\mathbf{z}_i}$. The composite node anomaly score combines attribute and structural residuals: $\mathcal{A}_i = w\|X_i - \hat{X}_i\|_2 + (1-w)\mathcal{A}_{\text{struct},i}$ ($w = 0.5$). During model training, summing squared errors over all $N$ nodes yields the global objective:

$$\mathcal{L}_{\text{struct}} = \frac{1}{N^2}\left[\|A\|_F^2 - 2\text{Tr}\left(\mathbf{Z}^\top A\mathbf{Z}\right) + \|\mathbf{Z}^\top \mathbf{Z}\|_F^2\right]. \tag{3}$$

In Equation (3), the Gram matrix $\mathbf{G} = \mathbf{Z}^\top \mathbf{Z}$ is of dimension $d \times d$ ($d \ll N$, typically $d = 64$), requiring only $O(d^2)$ memory. The sparse matrix-matrix multiplication $A\mathbf{Z}$ requires $O(|E|d)$ computation and storage. Thus, Equation (3) computes the exact mathematical loss with $O(Nd + |E|d + d^2)$ memory rather than $O(N^2)$. Numerical validation confirmed agreement with dense implementations within machine precision (error $< 10^{-7}$).

We emphasize that this exact algebraic reduction strictly requires an unsquashed dot-product decoder. For models such as AnomalyDAE that apply a non-linear sigmoid activation $\sigma(\mathbf{z}_i^\top \mathbf{z}_j)$ to structure reconstruction, the sum cannot be decoupled via Gram matrices in closed form. For AnomalyDAE, we instead implemented a chunked row-block evaluation that computes $B \times N$ cross-reconstructions iteratively ($B = 2,048$), reducing peak VRAM from $O(N^2)$ to $O(BN)$ while retaining exact $O(N^2)$ arithmetic.

## 4. Experimental Setup and Evaluation Protocol

### *4.1. Datasets and Provenance*

We evaluate 13 primary graph datasets spanning native financial ecosystems and standard benchmark graphs, plus one external enterprise cybersecurity network, as summarized in Table 1.

### *4.2. Evaluation Protocol and Metrics*

All models are evaluated transductively under five random seeds (seeds 42–46). Labels are strictly withheld during training.

- **Average Precision (PR-AUC):** Primary ranking metric, particularly sensitive to low-prevalence imbalances: $\text{AP} = \sum_k (R_k - R_{k-1})P_k$.
- **ROC-AUC:** Area under the receiver operating characteristic curve, measuring overall discrimination.
- **Alert-Budget Precision and Recall (P@K, R@K):** Evaluated at $K \in \{1\%, 5\%\}$ of total test nodes, measuring practical forensic yield when investigating top-ranked candidates.
- **Evaluation Prevalence Baseline:** The no-skill baseline $\pi_{\text{test}} = |V_{\text{test}}^+|/|V_{\text{test}}|$, against which AP gains must be judged.
- **Deterministic Split Identity:** All detectors on a given dataset and seed are evaluated on the exact same test node indices and labels. On DGraphFin, all six supported models evaluate the identical official 183,840-node test split (2,326 positives, $\pi_{\text{test}} = 0.0127$); the canonical run registry records two `split_hash` strings (`4ebf6ded...` for boolean-mask serialization vs. `3fd8c88f...` for flatnonzero index-array serialization in AnomalyDAE, verified as `SAME_MASK_DIFFERENT_METADATA`).

### *4.3. Friedman Hypothesis Testing and Permutation Design*

To assess statistical significance across multiple datasets without assuming normal distributions, we employ the Friedman non-parametric test. Let $r_i^j$ denote the rank of

**Table 1.** Benchmark graph dataset properties, evaluated model-input populations, target label provenance, and evaluation test prevalence. Model-input counts ($N, E, d$) reflect the exact tensors evaluated by all models; upstream raw source graph totals are reported separately. Graph Target Prevalence ($\pi_{\text{graph}} = y_{\text{pos}}/N$) reflects whole-graph label proportions; Evaluated Test Prevalence ($\pi_{\text{test}}$) defines the empirical no-skill AP baseline on the evaluated test split (mean across seeds 42–46).

| Dataset | Nodes ($N$) | Edges ($E$) | Dim ($d$) | $\pi_{\text{graph}}$ | $\pi_{\text{test}}$ | Target Provenance and Evaluated Population |
|---|---|---|---|---|---|---|
| *Native Financial / Blockchain Datasets ($N = 5$)* | | | | | | |
| Elliptic | 46,564 | 83,188 | 165 | 0.0976 | 0.0976 | Bitcoin transaction graph; illicit entities [34]. Evaluated induced graph: 46,564 nodes (raw source: 203,769 nodes). |
| DGraphFin | 1,225,601 | 746,271 | 17 | 0.0127 | 0.0127 | FinTech loan user graph; fraudulent defaulters [35]. Evaluated induced graph: 1,225,601 nodes (raw source: 3,700,550 nodes). |
| Ethereum | 14,385 | 115,065 | 8 | 0.4184 | 0.4089* | Token-transfer feature-similarity graph; verified phishing/scam contracts [36]. Test split: 2,893 nodes (40.89% fraud). |
| BSC | 7,481 | 59,832 | 8 | 0.1476 | 0.1473* | Binance Smart Chain token transfer graph; fraud contracts [36]. Test split: 1,500 nodes (14.73% fraud). |
| Polygon | 2,303 | 18,411 | 8 | 0.0261 | 0.0259* | Polygon token transfer graph; verified fraud contracts [36]. Test split: 471 nodes (2.59% fraud). |
| *Injected Synthetic Benchmarks ($N = 8$, including Real Trust Graph with Injected Targets)* | | | | | | |
| BitcoinOTC | 6,005 | 7,409 | 67 | 0.0300 | 0.0300 | SNAP OTC signed trust network [22,37]; controlled injected anomaly targets on real graph. Test split: 3,003 nodes. |
| Yelp-Syn | 716,847 | 14,004,995 | 300 | 0.0100 | 0.0100 | Social review network; injected structural and contextual anomalies. (PyG/PyGOD evaluated tensor: 716,847 nodes). Test split: 358,424 nodes. |
| Amazon-Syn | 13,752 | 507,378 | 767 | 0.0300 | 0.0300 | Co-purchasing graph; injected structural and contextual anomalies. (PyG/PyGOD evaluated tensor: 13,752 nodes). Test split: 6,876 nodes. |
| Reddit-Syn | 232,965 | 114,869,737 | 602 | 0.0200 | 0.0200 | Reddit post interaction graph; injected structural anomalies. (PyG/PyGOD evaluated tensor: 232,965 nodes). Test split: 116,483 nodes. |
| Flickr-Syn | 89,250 | 1,001,494 | 500 | 0.0200 | 0.0200 | Image sharing network; injected structural anomalies. (PyG/PyGOD evaluated tensor: 89,250 nodes). Test split: 44,625 nodes. |
| Cora-Syn | 2,708 | 11,276 | 1433 | 0.0299 | 0.0298* | Citation network; injected structural anomalies (PyGOD generator). (PyG/PyGOD evaluated tensor: 2,708 nodes). Test split: 1,354 nodes. |
| CiteSeer-Syn | 3,327 | 9,914 | 3703 | 0.0298 | 0.0300 | Citation network; injected structural anomalies (PyGOD generator). (PyG/PyGOD evaluated tensor: 3,327 nodes). Test split: 1,664 nodes. |
| PubMed-Syn | 19,717 | 93,958 | 500 | 0.0300 | 0.0300 | Citation network; injected structural anomalies (PyGOD generator). (PyG/PyGOD evaluated tensor: 19,717 nodes). Test split: 9,859 nodes. |
| *External Enterprise Cybersecurity Network ($N = 1$)* | | | | | | |
| LANL-RedTeam | 16,694 | 323,897 | 13 | 0.0180 | 0.0180 | Enterprise authentication graph; red-team compromised computers [38]. Evaluated destination computers: 16,694 nodes. Test split: 3,338 nodes. |

* For datasets with random stratified cross-seed splits, $\pi_{\text{test}}$ denotes the empirical mean test-set positive rate across seeds 42–46 (Ethereum: 0.3951–0.4307, mean 0.4089; BSC: 0.1353–0.1600, mean 0.1473; Polygon: 0.0149–0.0382, mean 0.0259; Cora: 0.0295–0.0303, mean 0.0298).

detector $j \in \{1, \ldots, k\}$ on dataset block $i \in \{1, \ldots, n\}$. The Friedman statistic $Q$ is computed with tie corrections:

$$Q = \frac{\frac{12}{nk(k+1)} \sum_{j=1}^{k} R_j^2 - 3n(k+1)}{1 - \frac{\sum T_i}{n(k^3-k)}}, \tag{4}$$

where $R_j = \sum_{i=1}^{n} r_i^j$ is the sum of ranks for model $j$.

While $Q$ converges asymptotically to $\chi^2_{k-1}$ as $n \to \infty$, benchmark suites typically have small sample sizes ($n \in \{6, 8, 13\}$). Under small $n$, the continuous $\chi^2$ approximation tends to be conservative. We therefore compute exact Monte Carlo within-block permutation $p$-values: within each dataset row, the model ranks are permuted randomly across $B = 1,000,000$ replications (pinned seed 20261024), computing the empirical tail probability $p = (b+1)/(B+1)$ where $b$ is the number of permutations yielding a statistic greater than or equal to $Q_{\text{obs}}$. For pairwise comparisons against DLG-Aug, we report two-sided Wilcoxon signed-rank tests adjusted via the Holm step-down procedure [39]. We note that for within-dataset comparisons across $N_{\text{seeds}} = 5$, the minimum achievable two-sided Wilcoxon signed-rank $p$-value is $2^{-(5-1)} = 0.0625$ when all five differences share the same nonzero sign. Consequently, no single dataset comparison can attain conventional $\alpha = 0.05$ significance under five seeds alone; statistical hypothesis testing is therefore evaluated across the benchmark suite via the Friedman permutation framework. We note that for within-dataset comparisons across $N_{\text{seeds}} = 5$, the minimum achievable two-sided Wilcoxon signed-rank $p$-value is $2^{-(5-1)} = 0.0625$ when all five differences share the same nonzero sign. Consequently, no single dataset comparison can attain conventional $\alpha = 0.05$ significance under five seeds alone; statistical hypothesis testing is therefore evaluated across the benchmark suite via the Friedman permutation framework.

## 5. Results and Empirical Analysis

### *5.1. Evaluated Population and Hardware Support Feasibility*

Figure 2 details the execution feasibility across all 14 datasets and seven detector configurations.

Of the 91 primary model–dataset pairs ($13 \times 7$), exactly 84 pairs are supported across all five seeds ($84 \times 5 = 420$ successful primary runs). As recorded in `support_14x7.csv`, GADNR fails on five graphs across two distinct categories: four `NO_APPROVED_RAW_RUN` pairs (Elliptic, DGraphFin, Yelp-Syn, and Reddit-Syn; historical execution failure / numerical covariance determinant limitation) and one `OBSERVED_HISTORICAL_CUDA_OOM` pair (Flickr-Syn; historical CUDA OOM on an 8-GB legacy environment). AnomalyDAE fails on two graphs (Yelp-Syn and Reddit-Syn) with `OBSERVED_CUDA_OOM` from preflight CUDA `OutOfMemoryError` exceptions in the full encoder forward pass on the 24-GB RTX 3090 (seven unsupported primary pairs in total). In contrast, the exact Gram formulations developed in Section 3 allow DOMINANT, DLG-Base, and DLG-Aug to complete all runs successfully without memory overflow, even on the 1.2-million-node DGraphFin graph.

### *5.2. Predictive Quality vs. Evaluation Prevalence and Alert Budgets*

Table 2 reports the Average Precision (PR-AUC) performance of all seven detectors alongside the evaluation prevalence baseline.

**Interpretation of Prevalence Baselines and Comparator Superiority:**

1. On the Elliptic dataset, DLG-Aug is the only model achieving AP higher than the test prevalence ($0.1037 > 0.0976$), whereas DOMINANT (0.0682), AnomalyDAE (0.0538), and DLG-Base (0.0687) fall below random baseline precision.

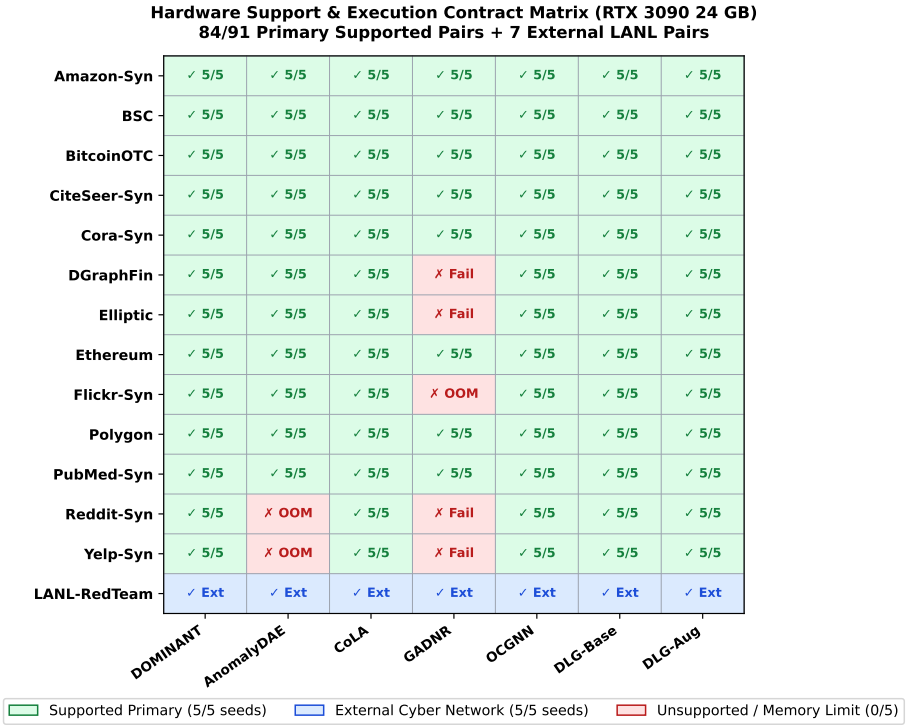



**Figure 2.** Hardware support feasibility and failure provenance matrix across 14 graph datasets and seven models (84 of 91 primary pairs fully supported across all five seeds). Cell annotations and shading distinguish exact complete primary support (`5/5`), external benchmark support on LANL-RedTeam (`Ext`), historical execution or numerical covariance determinant failure without an approved raw run (`Fail`: four GADNR pairs on Elliptic, DGraphFin, Yelp-Syn, and Reddit-Syn), and out-of-memory hardware exhaustion (`OOM`: one historical 8-GB legacy CUDA OOM for GADNR on Flickr-Syn, and two preflight 24-GB RTX 3090 full-encoder-forward CUDA OOMs for AnomalyDAE on Yelp-Syn and Reddit-Syn).

**Table 2.** Average Precision (PR-AUC; mean across five seeds). Evaluated test prevalence ($\pi_{\text{test}}$) defines the empirical no-skill random-ranking baseline on the evaluated test split. Unsupported cells are denoted as N/A.

| Dataset | $\pi_{\text{test}}$ | DOMINANT | AnomalyDAE | CoLA | GADNR | OCGNN | DLG-Base | DLG-Aug |
|---|---|---|---|---|---|---|---|---|
| Elliptic | 0.0976 | 0.0682 | 0.0538 | 0.0947 | N/A | 0.0667 | 0.0687 | **0.1037** |
| DGraphFin | 0.0127 | 0.0102 | 0.0119 | 0.0127 | N/A | 0.0126 | 0.0101 | **0.0134** |
| BitcoinOTC | 0.0300 | **0.8810** | 0.7948 | 0.4936 | 0.8125 | 0.0290 | 0.8274 | 0.8211 |
| Ethereum | 0.4089 | 0.3971 | 0.3986 | 0.3648 | 0.3976 | **0.4996** | 0.4106 | 0.4359 |
| BSC | 0.1473 | 0.2299 | **0.3609** | 0.1252 | 0.1561 | 0.1615 | 0.3390 | 0.3411 |
| Polygon | 0.0259 | 0.1071 | **0.1878** | 0.0232 | 0.0260 | 0.0700 | 0.0854 | 0.0731 |
| Yelp-Syn | 0.0100 | 0.0749 | N/A | 0.0174 | N/A | 0.0069 | 0.0826 | **0.0832** |
| Amazon-Syn | 0.0300 | 0.1301 | **0.2734** | 0.0989 | 0.0338 | 0.0249 | 0.1245 | 0.1302 |
| Reddit-Syn | 0.0200 | **0.4069** | N/A | 0.1606 | N/A | 0.0205 | 0.4044 | 0.3388 |
| Flickr-Syn | 0.0200 | 0.4610 | **0.4761** | 0.0447 | N/A | 0.0247 | 0.4569 | 0.4599 |
| Cora-Syn | 0.0298 | 0.4740 | 0.4494 | 0.0299 | 0.0951 | 0.0302 | **0.4758** | 0.4720 |
| CiteSeer-Syn | 0.0300 | 0.5989 | 0.5507 | 0.0304 | 0.3662 | 0.0301 | **0.6004** | 0.5992 |
| PubMed-Syn | 0.0300 | 0.1607 | **0.6837** | 0.0705 | 0.1014 | 0.0300 | 0.1623 | 0.1605 |
| LANL-RedTeam | 0.0180 | 0.1348 | 0.0450 | 0.0200 | **0.1806** | 0.0454 | 0.1367 | 0.1114 |

Note: For datasets with random stratified cross-seed splits (Ethereum, BSC, Polygon, Cora), $\pi_{\text{test}}$ reports the empirical mean test positive rate across seeds 42–46.

2. On DGraphFin, DLG-Aug (0.0134) and CoLA (0.0127) perform at or above the verified evaluation prevalence baseline (0.0127), while other baselines perform near or below prevalence.
3. On the three cryptocurrency networks, established comparators outperform DLG-Aug: OCGNN achieves the highest AP on Ethereum (0.4996 vs. Aug 0.4359, both exceeding the 0.4089 test prevalence), while AnomalyDAE leads on BSC (0.3609 vs. Aug 0.3411) and Polygon (0.1878 vs. Aug 0.0731). Thus, *at least one established comparator exceeds DLG-Aug in mean AP on each of the three blockchain datasets.*

Table 3 evaluates operational alert budgets on the three cryptocurrency graphs, measuring Precision@1%, Recall@1%, Precision@5%, and Recall@5%.

**Table 3.** Operational Alert-Budget Performance across Cryptocurrency Benchmarks (mean across five seeds). Metrics denote precision and recall when auditing the top-1% and top-5% highest-scoring nodes.

| Dataset | Metric | DOMINANT | AnomalyDAE | CoLA | GADNR | OCGNN | DLG-Base | DLG-Aug |
|---|---|---|---|---|---|---|---|---|
| Ethereum | *P*@1% | **0.4897** | 0.2621 | 0.3034 | 0.4621 | 0.3931 | 0.4690 | 0.4621 |
| | *R*@1% | **0.0121** | 0.0064 | 0.0074 | 0.0113 | 0.0098 | 0.0115 | 0.0114 |
| | *P*@5% | **0.5145** | 0.3117 | 0.3779 | 0.4110 | 0.5090 | 0.4800 | 0.5007 |
| | *R*@5% | 0.0631 | 0.0381 | 0.0463 | 0.0503 | **0.0632** | 0.0591 | 0.0620 |
| BSC | *P*@1% | 0.1867 | **0.3867** | 0.0533 | 0.1733 | 0.2133 | 0.2800 | 0.3067 |
| | *R*@1% | 0.0126 | **0.0261** | 0.0036 | 0.0116 | 0.0145 | 0.0187 | 0.0207 |
| | *P*@5% | 0.2827 | **0.4507** | 0.0720 | 0.1653 | 0.1600 | 0.4133 | 0.4133 |
| | *R*@5% | 0.0959 | **0.1527** | 0.0242 | 0.0556 | 0.0545 | 0.1397 | 0.1402 |
| Polygon | *P*@1% | 0.2400 | **0.3600** | 0.0000 | 0.0000 | 0.0400 | 0.0400 | 0.0800 |
| | *R*@1% | 0.1114 | **0.1566** | 0.0000 | 0.0000 | 0.0286 | 0.0111 | 0.0236 |
| | *P*@5% | 0.1083 | **0.1500** | 0.0000 | 0.0000 | 0.0583 | 0.1000 | 0.0833 |
| | *R*@5% | 0.2102 | **0.2964** | 0.0000 | 0.0000 | 0.1323 | 0.1984 | 0.1651 |

### *5.3. Conditional Utility Analysis and Delta Distribution*

Figure 3 and Table 4 illustrate the signed performance differential ($\Delta AP = AP_{Aug} - AP_{Base}$) across all 14 evaluated benchmarks.

The empirical evidence disproves any universal benefit from pretrained local-feature augmentation. While DLG-Aug yields valuable improvements on transaction graphs like Elliptic ($\Delta AP = +0.0350$) and Ethereum ($\Delta AP = +0.0253$), it exhibits lower performance on Reddit-Syn ($\Delta AP = -0.0656$) and LANL-RedTeam ($\Delta AP = -0.0252$).

### *5.4. Cross-Dataset Omnibus Evaluation and Permutation Statistics*

Table 5 summarizes the Friedman omnibus test results across four evaluation views (S1–S4).

**Statistical Explanation of Asymptotic vs. Permutation Discrepancies:** The asymptotic $\chi^2$ approximation and the within-block Monte Carlo permutation test yielded different tail probabilities for the smaller complete-case views (S1, S2, and S4). We therefore report the permutation design, random seed (seed 20261024), replicate count ($B = 1{,}000{,}000$), exceedance counts, and Monte Carlo uncertainty alongside the asymptotic reference values:

- When $n$ is small ($n \leq 8$), the discrete permutation space $\prod_{i=1}^{n}(k!)$ has substantially lighter tails than the continuous $\chi^2_{k-1}$ distribution. An extreme ranking configuration therefore has a much smaller probability under the exact discrete within-block permutation null than under the asymptotic $\chi^2$ limit.
- At $B = 1{,}000{,}000$, we observe exactly 3 exceedances for S1 ($p = 4.0 \times 10^{-6}$), 20 for S2 ($p = 2.1 \times 10^{-5}$), and 0 for S4 ($p = 1.0 \times 10^{-6}$, finite-simulation estimator floor

**Table 4.** Conditional Average Precision performance of DLG-Base versus DLG-Aug across all 14 evaluated datasets (13 primary graphs plus external LANL-RedTeam). Results report mean AP and seed-paired delta ($\Delta AP = AP_{Aug} - AP_{Base}$) alongside paired standard errors ($n = 5$ random seeds). Localized structural augmentation provides clear utility in sparse transaction graphs but exhibits performance trade-offs on dense social networks and cyber authentication graphs.

| Dataset | DLG-Base (AP) | DLG-Aug (AP) | ΔAP | Paired SE | Operational Characterization |
|---|---|---|---|---|---|
| Elliptic | 0.0687 | 0.1037 | +**0.0350** | ±0.0022 | **Large Gain** (+50.97% rel.); positive ΔAP observed; mechanism not isolated by this comparison. |
| DGraphFin | 0.0101 | 0.0134 | +0.0034 | ±0.0008 | **Gain** (+33.23% rel.); exact full-graph execution supported under the reported envelope. |
| BitcoinOTC | 0.8274 | 0.8211 | −0.0063 | ±0.0033 | **Slight Loss** (−0.77% rel.); high baseline precision (0.827) retains competitive performance. |
| Ethereum | 0.4106 | 0.4359 | +**0.0253** | ±0.0144 | **Gain** (+6.15% rel.); DLG-Aug has higher mean AP in this constructed similarity-graph evaluation. |
| BSC | 0.3390 | 0.3411 | +0.0021 | ±0.0103 | **Marginal Gain** (+0.61% rel.); slight positive mean difference on BSC token transfers. |
| Polygon | 0.0854 | 0.0731 | −0.0124 | ±0.0054 | **Loss** (−14.48% rel.); negative ΔAP observed in this similarity-graph evaluation. |
| Yelp-Syn | 0.0826 | 0.0832 | +0.0007 | ±0.0015 | Stable / marginal gain on injected contextual/structural anomaly patterns. |
| Amazon-Syn | 0.1245 | 0.1302 | +0.0057 | ±0.0016 | Stable / marginal gain on injected contextual/structural anomaly patterns. |
| Reddit-Syn | 0.4044 | 0.3388 | −**0.0656** | ±0.0031 | **Substantial Loss** (−16.22% rel.); negative delta observed on dense social interaction graph. |
| Flickr-Syn | 0.4569 | 0.4599 | +0.0030 | ±0.0076 | Stable / marginal gain on injected contextual/structural anomaly patterns. |
| Cora-Syn | 0.4758 | 0.4720 | −0.0038 | ±0.0031 | Stable citation network performance; local features preserve structural fidelity. |
| CiteSeer-Syn | 0.6004 | 0.5992 | −0.0013 | ±0.0014 | Stable citation network performance; local features preserve structural fidelity. |
| PubMed-Syn | 0.1623 | 0.1605 | −0.0018 | ±0.0003 | Stable citation network performance; local features preserve structural fidelity. |
| LANL-RedTeam† | 0.1367 | 0.1114 | −0.0252 | ±0.0126 | **External Loss** (−18.47% rel.); negative ΔAP observed on cyber authentication event graph. |

† LANL-RedTeam is an external enterprise cybersecurity network evaluated under identical protocol but reported as single-graph descriptive evidence.

**Table 5.** Complete-case Friedman non-parametric test statistics and $p$-values across four evaluation views under 1,000,000 Monte Carlo within-block permutations (seed 20261024). $n$ and $k$ denote the number of dataset blocks and comparative models, respectively; $Q$ is the Friedman test statistic; $b$ is the permutation exceedance count. Exact tail probabilities are computed as $p = (b + 1)/(B + 1)$ alongside asymptotic $\chi^2$ references.

| View | $n$ | $k$ | Stat ($Q$) | $\chi^2$ $p$ | Exceed. ($b$) | 1M MC $p \pm$ SE | Statistical Inference |
|---|---|---|---|---|---|---|---|
| **S1** (8 DS, 7 M) | 8 | 7 | 29.0357 | $5.99e-05$ | 3 | $4.00e-06 \pm 2.0e-06$ | **Significant** ($p < 0.05$); reject null of equal detector ranking. |
| **S2** (13 DS, 5 M) | 13 | 5 | 23.7538 | $8.95e-05$ | 20 | $2.10e-05 \pm 4.6e-06$ | **Significant** ($p < 0.05$); reject null of equal detector ranking. |
| **S3** (6 DS, 5 M) | 6 | 5 | 6.1333 | $1.89e-01$ | 192,107 | $1.92e-01 \pm 3.9e-04$ | *Non-significant* ($p = 0.1921$); null of equal ranking retained. |
| **S4** (7 DS, 5 M) | 7 | 5 | 21.8286 | $2.17e-04$ | 0 | $1.00e-06 \pm 1.0e-06$ | **Significant** ($p < 0.05$); reject null of equal detector ranking. |

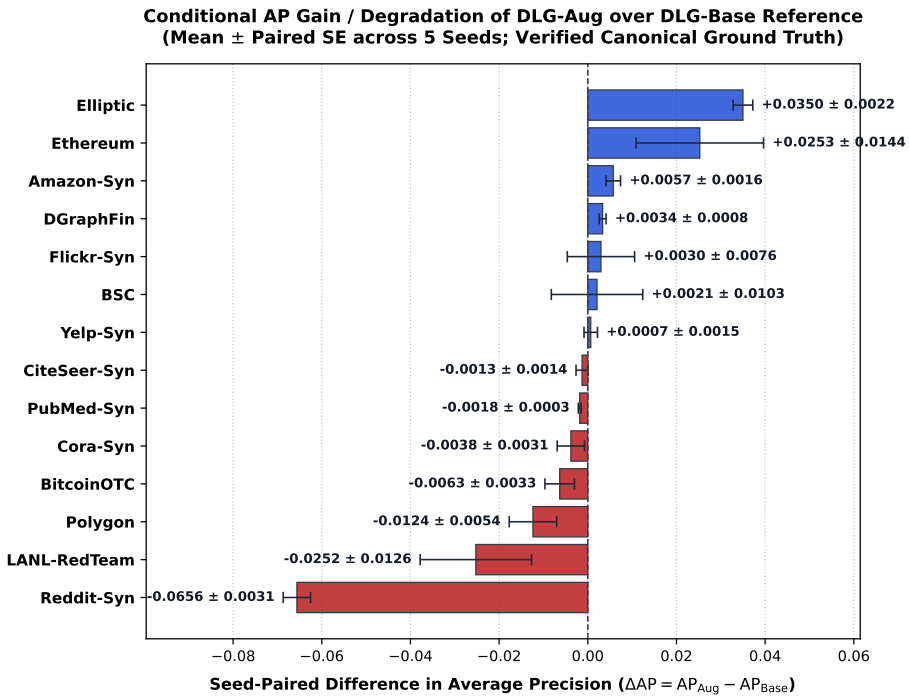



**Figure 3.** Conditional AP delta ($\Delta\text{AP} = \text{AP}_{\text{Aug}} - \text{AP}_{\text{Base}}$) across all 13 primary datasets plus external LANL-RedTeam (with paired seed-level standard error bars, $n = 5$ seeds). Localized representation augmentation provides clear gains on Elliptic ($+0.0350$) and Ethereum ($+0.0253$), marginal gain on BSC ($+0.0021$), but incurs performance degradation on Polygon ($-0.0124$), Reddit-Syn ($-0.0656$), and LANL-RedTeam ($-0.0252$).

bounded by $-\ln(0.05)/B \approx 3.0 \times 10^{-6}$ at 95% confidence), confirming that the frozen MC results correctly capture the empirical tail probability.

Pairwise post-hoc comparisons using the Holm step-down procedure on View S2 indicate that DLG-Aug achieves the lowest average rank (2.0000), followed closely by DOMINANT (2.3846) and DLG-Base (2.5385). However, adjusted Wilcoxon tests between DLG-Aug and DOMINANT ($p_{\text{adj}} = 1.0000$) and DLG-Base ($p_{\text{adj}} = 1.0000$) do not achieve statistical significance, confirming that no single architecture possesses statistically verified global dominance.

## 6. Execution, Memory, and Computational Complexity

To illustrate the practical computational scaling of the evaluated methods, Table 6 compares theoretical asymptotic complexity against illustrative peak GPU memory materialization profiles on an NVIDIA GeForce RTX 3090 24 GB GPU platform.

The closed-form Gram formulation reduces the asymptotic space complexity of DOMINANT and DLG models from quadratic $\mathcal{O}(N^2)$ to linear in edge density $\mathcal{O}(|E|d + Nd)$. On Reddit-Syn ($N = 232,965$, $|E| = 114.9\,\text{M}$ directed edges), this allows full-graph training within 5.5 GB of VRAM, whereas standard implementations crash with out-of-memory errors. Importantly, row-blocked AnomalyDAE alleviates memory footprint ($\mathcal{O}(BN)$) but retains $\mathcal{O}(N^2)$ arithmetic operations, demonstrating the distinction between memory scalability and runtime complexity.

**Table 6.** Theoretical computational complexity and illustrative peak memory materialization profiles for graph anomaly detector architectures. Memory entries indicate estimated peak allocated GPU memory scaling profiles under PyTorch (batch size $B = 2{,}048$ for row-blocked formulations) on an RTX 3090 24GB platform; values serve as illustrative architectural scaling profiles rather than hardware-invariant empirical benchmarks. On Reddit-Syn, full AnomalyDAE remains unsupported in the benchmark due to dense structure encoder memory limits.

| Model | Time Complexity | Space Complexity | Profile (Cora) | Profile (Reddit) |
|---|---|---|---|---|
| DOMINANT (Dense PyGOD) | $\mathcal{O}(L\|E\|d + N^2d)$ | $\mathcal{O}(N^2 + Nd)$ | $\approx 1.2$ GB | OOM ($> 24$ GB) |
| DOMINANT (Exact Gram) | $\mathcal{O}(L\|E\|d + Nd^2)$ | $\mathcal{O}(\|E\|d + Nd + d^2)$ | $\approx 0.4$ GB | $\approx 4.8$ GB |
| AnomalyDAE (Dense) | $\mathcal{O}(L\|E\|d + N^2d)$ | $\mathcal{O}(N^2 + Nd)$ | $\approx 1.9$ GB | OOM ($> 24$ GB) |
| AnomalyDAE (Row-Block)$^\dagger$ | $\mathcal{O}(L\|E\|d + N^2d)$ | $\mathcal{O}(BN + Nd)$ | $\approx 0.7$ GB | $\approx 7.1$ GB$^\dagger$ |
| CoLA | $\mathcal{O}(SK\|E_{\text{sub}}\|d)$ | $\mathcal{O}(SKN_{\text{sub}}d)$ | $\approx 0.8$ GB | $\approx 5.9$ GB |
| GADNR | $\mathcal{O}(L\|E\|d + NKd^2)$ | $\mathcal{O}(\|E\|d + NKd)$ | $\approx 1.5$ GB | OOM ($> 24$ GB) |
| OCGNN | $\mathcal{O}(L\|E\|d + Nd)$ | $\mathcal{O}(\|E\|d + Nd)$ | $\approx 0.3$ GB | $\approx 2.5$ GB |
| DLG-Base (Exact Gram) | $\mathcal{O}(L\|E\|d + Nd^2)$ | $\mathcal{O}(\|E\|d + Nd + d^2)$ | $\approx 0.4$ GB | $\approx 4.9$ GB |
| DLG-Aug (Exact Gram) | $\mathcal{O}((L_{\text{loc}} + L_{\text{glob}})\|E\|d)$ | $\mathcal{O}(\|E\|d + Nd + d^2)$ | $\approx 0.5$ GB | $\approx 5.5$ GB |

## 7. Discussion: Practical Anomaly-Detection Deployment Guide

### *7.1. Practitioner Decision Matrix*

Based on our multi-dataset evaluation, Table 7 presents an operational decision matrix providing conditional starting points for practitioners deploying unsupervised GAD in financial and cyber domains.

### *7.2. Limitations and Boundary Conditions*

We explicitly delineate the scientific boundaries of this study:

1. **Static vs. Dynamic Formulations:** Although cryptocurrency transactions and authentication logs occur continuously over time, our benchmark evaluates frozen static projections to ensure comparability across standard GAD architectures. Dynamic graph models and continuous-time temporal architectures (e.g., TGN [40], JODIE [41], TGAT [42], EvolveGCN [43], DyRep [44]) remain valuable future benchmarks.
2. **Constructed Similarity Adjacency in Crypto Graphs:** Blockchain networks (Ethereum, BSC, Polygon) utilize feature-similarity graphs constructed from observable transfer histories rather than direct payment graphs, reflecting contract-level similarity rather than money flows.
3. **Hardware Specificity:** Memory measurements reflect PyTorch 2.14 / CUDA 13.0 execution on an NVIDIA GeForce RTX 3090 24 GB GPU connected via a Thunderbolt 3 eGPU enclosure to host notebook (Intel Core i9-13900H / RTX 4070 laptop GPU); absolute allocations may vary across operating systems or multi-GPU environments.

## 8. Conclusion

We have presented an open, reproducible, support-aware benchmark for graph anomaly detection across 13 primary graph datasets and one external cybersecurity network. By establishing a rigorous evaluation protocol that separates execution feasibility from predictive precision, we demonstrate that unsupervised GAD performance is highly domain-dependent. Decoupled local-to-global augmentation provides distinct advantages on transaction graphs such as Elliptic ($+0.0350$ AP gain), whereas standard autoencoders obtain below-random ROC-AUC under standard scoring, but it incurs performance trade-offs on dense social networks and cyber authentication graphs. Furthermore, our closed-form Gram decoders establish that exact full-graph reconstruction is achievable without dense $N \times N$ memory allocation for unsquashed dot-product decoders. This benchmark and

**Table 7.** Practitioner decision matrix providing conditional starting points for detector selection under operational constraints.

| Operating Condition | Recommended Model | Secondary Alternative | Unsuitable Models |
|---|---|---|---|
| Transaction networks with extreme imbalance (e.g., Elliptic) | **DLG-Aug** (achieves highest AP, $\Delta$AP $= +0.0350$ absolute uplift over DLG-Base, surpassing test prevalence $0.1037 > 0.0976$) | **CoLA** (contrastive local consensus) | DOMINANT, AnomalyDAE (exhibit below-random ROC-AUC under standard orientation) |
| Large-scale transaction graphs ($N > 10^6$, low memory budget) | **DLG-Base / DOMINANT** with Exact Gram decoder | **OCGNN** (lightweight SVDD objective) | GADNR (covariance/memory limits on large graphs), dense autoencoders |
| Dense interaction graphs (e.g., Reddit-Syn) | **DOMINANT / DLG-Base** | **AnomalyDAE** (with row-blocking) | DLG-Aug (exhibits lower AP than DLG-Base on dense community graphs; $\Delta$AP $= -0.0656$) |
| Strict Alert Budgets ($P$@1% prioritization in AML triage) | **AnomalyDAE** (leads Precision@1% on BSC and Polygon) | **DOMINANT / DLG-Aug** (DOMINANT leads on Ethereum; DLG-Aug competitive across all three) | CoLA, GADNR (insufficient top-rank precision on low-prevalence chains; descriptive of tested graph settings) |
| Cyber authentication logs (e.g., LANL-RedTeam) | **GADNR** (Gaussian mixture neighborhood reconstruction) | **DLG-Base** | DLG-Aug (exhibits negative delta on authentication event graphs; $\Delta$AP $= -0.0252$) |

decision guide provide practitioners with a sound foundation for selecting and deploying graph anomaly detection systems under realistic operational constraints.

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# Appendix: Supplementary Experimental Tables

**Table A1.** Full Area Under the ROC Curve (ROC-AUC; mean ± SD across five seeds). Values below 0.50 reflect empirical structural behavior under standard detector orientation.

| Dataset | DOMINANT | AnomalyDAE | CoLA | GADNR | OCGNN | DLG-Base | DLG-Aug |
|---|---|---|---|---|---|---|---|
| Elliptic | 0.3432 ± 0.0367 | 0.1296 ± 0.0246 | 0.4878 ± 0.0088 | N/A | 0.3140 ± 0.0466 | 0.3500 ± 0.0221 | **0.5705 ± 0.0228** |
| DGraphFin | 0.4088 ± 0.0049 | 0.4691 ± 0.0971 | 0.4994 ± 0.0088 | N/A | 0.4951 ± 0.0486 | 0.4092 ± 0.0086 | **0.5185 ± 0.0364** |
| BitcoinOTC | **0.9882 ± 0.0080** | 0.9671 ± 0.0214 | 0.8035 ± 0.0169 | 0.9237 ± 0.1301 | 0.4926 ± 0.0838 | 0.9874 ± 0.0056 | 0.9875 ± 0.0071 |
| Ethereum | 0.4336 ± 0.0284 | 0.4734 ± 0.0272 | 0.4212 ± 0.0078 | 0.4788 ± 0.0165 | **0.6007 ± 0.1492** | 0.4867 ± 0.0272 | 0.5326 ± 0.0355 |
| BSC | 0.6464 ± 0.0337 | **0.7743 ± 0.0160** | 0.4386 ± 0.0314 | 0.5010 ± 0.0209 | 0.5295 ± 0.0719 | 0.7620 ± 0.0215 | 0.7551 ± 0.0165 |
| Polygon | 0.6283 ± 0.0653 | **0.7977 ± 0.0301** | 0.4260 ± 0.0533 | 0.4637 ± 0.1072 | 0.5498 ± 0.0761 | 0.7660 ± 0.0463 | 0.7116 ± 0.0382 |
| Yelp-Syn | 0.9348 ± 0.0011 | N/A | 0.4250 ± 0.0803 | N/A | 0.3461 ± 0.0283 | 0.9385 ± 0.0012 | **0.9400 ± 0.0014** |
| Amazon-Syn | **0.8123 ± 0.0176** | 0.6862 ± 0.0857 | 0.6544 ± 0.0421 | 0.5532 ± 0.0098 | 0.4242 ± 0.0187 | 0.8098 ± 0.0175 | 0.8104 ± 0.0181 |
| Reddit-Syn | **0.9700 ± 0.0006** | N/A | 0.7657 ± 0.0256 | N/A | 0.5155 ± 0.0047 | 0.9690 ± 0.0004 | 0.9657 ± 0.0006 |
| Flickr-Syn | 0.9813 ± 0.0006 | **0.9814 ± 0.0039** | 0.3294 ± 0.1078 | N/A | 0.5014 ± 0.0764 | 0.9810 ± 0.0010 | 0.9812 ± 0.0005 |
| Cora-Syn | 0.9844 ± 0.0017 | 0.9619 ± 0.0037 | 0.5078 ± 0.0133 | 0.6800 ± 0.1513 | 0.4916 ± 0.0075 | **0.9845 ± 0.0016** | 0.9842 ± 0.0017 |
| CiteSeer-Syn | **0.9780 ± 0.0057** | 0.9763 ± 0.0052 | 0.4962 ± 0.0209 | 0.9185 ± 0.0297 | 0.4994 ± 0.0019 | 0.9777 ± 0.0058 | 0.9779 ± 0.0058 |
| PubMed-Syn | 0.9265 ± 0.0026 | **0.9886 ± 0.0002** | 0.5560 ± 0.0232 | 0.8222 ± 0.1196 | 0.5000 ± 0.0004 | 0.9272 ± 0.0027 | 0.9264 ± 0.0027 |
| LANL-RedTeam | 0.7997 ± 0.0237 | 0.5548 ± 0.0502 | 0.4663 ± 0.0687 | **0.8261 ± 0.0195** | 0.3762 ± 0.0945 | 0.7923 ± 0.0404 | 0.7188 ± 0.0904 |

**Table A2.** Validation-thresholded Test F1 Score (mean ± SD across five seeds). Thresholds selected solely on validation splits.

| Dataset | DOMINANT | AnomalyDAE | CoLA | GADNR | OCGNN | DLG-Base | DLG-Aug |
|---|---|---|---|---|---|---|---|
| Elliptic | 0.1850 ± 0.0060 | 0.1778 ± 0.0001 | 0.1782 ± 0.0003 | N/A | 0.1796 ± 0.0026 | 0.1817 ± 0.0017 | **0.2239 ± 0.0164** |
| DGraphFin | 0.0250 ± 0.0000 | 0.0286 ± 0.0053 | 0.0253 ± 0.0006 | N/A | 0.0279 ± 0.0038 | 0.0250 ± 0.0000 | **0.0296 ± 0.0059** |
| BitcoinOTC | **0.9329 ± 0.0060** | 0.8385 ± 0.0397 | 0.5400 ± 0.0240 | 0.8675 ± 0.2276 | 0.0788 ± 0.0119 | 0.9256 ± 0.0027 | 0.9260 ± 0.0081 |
| Ethereum | 0.5829 ± 0.0179 | 0.5837 ± 0.0177 | 0.5802 ± 0.0142 | 0.5797 ± 0.0169 | **0.6437 ± 0.0560** | 0.5908 ± 0.0099 | 0.5963 ± 0.0137 |
| BSC | 0.3150 ± 0.0371 | 0.4592 ± 0.0351 | 0.2582 ± 0.0174 | 0.2529 ± 0.0157 | 0.2776 ± 0.0241 | **0.4660 ± 0.0234** | 0.4534 ± 0.0290 |
| Polygon | **0.2216 ± 0.0802** | 0.2105 ± 0.0645 | 0.0277 ± 0.0267 | 0.0188 ± 0.0184 | 0.0397 ± 0.0418 | 0.1292 ± 0.0403 | 0.0756 ± 0.0489 |
| Yelp-Syn | 0.1505 ± 0.0030 | N/A | 0.0573 ± 0.0093 | N/A | 0.0200 ± 0.0003 | 0.1630 ± 0.0025 | **0.1669 ± 0.0047** |
| Amazon-Syn | 0.2636 ± 0.0141 | **0.3332 ± 0.1841** | 0.1630 ± 0.0360 | 0.0650 ± 0.0026 | 0.0575 ± 0.0009 | 0.2513 ± 0.0153 | 0.2633 ± 0.0194 |
| Reddit-Syn | **0.4685 ± 0.0083** | N/A | 0.2806 ± 0.0596 | N/A | 0.0406 ± 0.0006 | 0.4642 ± 0.0055 | 0.4289 ± 0.0069 |
| Flickr-Syn | 0.5162 ± 0.0050 | **0.5303 ± 0.0195** | 0.1141 ± 0.0454 | N/A | 0.0605 ± 0.0137 | 0.5138 ± 0.0070 | 0.5110 ± 0.0053 |
| Cora-Syn | 0.6852 ± 0.0262 | 0.4399 ± 0.0594 | 0.0450 ± 0.0180 | 0.1597 ± 0.2127 | 0.0585 ± 0.0008 | **0.6892 ± 0.0268** | 0.6865 ± 0.0279 |
| CiteSeer-Syn | 0.7489 ± 0.0109 | 0.5618 ± 0.0465 | 0.0412 ± 0.0178 | 0.5422 ± 0.1778 | 0.0584 ± 0.0001 | **0.7502 ± 0.0098** | 0.7489 ± 0.0109 |
| PubMed-Syn | 0.3543 ± 0.0045 | **0.6779 ± 0.0064** | 0.1111 ± 0.0172 | 0.2455 ± 0.1119 | 0.0583 ± 0.0000 | 0.3561 ± 0.0056 | 0.3543 ± 0.0045 |
| LANL-RedTeam | 0.1740 ± 0.0637 | 0.0688 ± 0.0417 | 0.0295 ± 0.0149 | **0.2475 ± 0.0519** | 0.0568 ± 0.0125 | 0.2112 ± 0.0254 | 0.1683 ± 0.0458 |